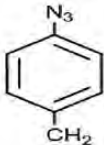
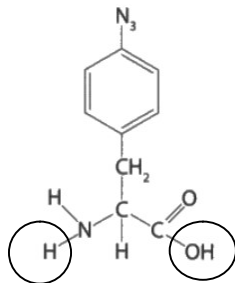


Question number	Answer	Mark
8(a)	The only correct answer is B.  <i>A is incorrect because the R group does not include the central carbon</i> <i>C is incorrect because the R group should have a CH₂ group</i> <i>D is incorrect because the H and central carbon and only part of the R group shown</i>	(1)

Question number	Answer	Additional guidance	Mark
8(b)(i)	peptide / amide (bond)	DO NOT ACCEPT dipeptide / polypeptide (bond)	(1)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	A drawing that shows : - a circle around one H on the N (1) - a circle around only the OH group on the C (1)		(2)

Question number	Answer	Additional guidance	Mark
8(b)(iii)	An explanation that includes three of the following points : • because AzPhe is not a naturally-occurring amino acid (1) • therefore there is no {DNA / mRNA} codon for AzPhe (1) • therefore there is no tRNA (that can bind to the AzPhe) (1) • therefore AzPhe not held in position for peptide bond to form (1)		(3)

Question number	Answer	Additional guidance	Mark
8(c)(i)	A calculation showing the following steps: • the percentage of AzPhe are calculated (1) • ratio calculated (1)	Example of calculation : 16% of 1% of 1 100 = 1.76 5.25 : 1 / 21 : 4 ACCEPT 1 : 0.19 Correct answer with no working shown gains both marks.	(2)

Question number	Answer	Additional guidance	Mark
8(c)(ii)	An answer that includes four of the following points: - properties of a protein is dependent on the structure of a protein (1) - AzPhe could affect the {secondary structure / folding / 3D shape} (of the silk) (1) - because different bonds (between R groups) could form (1) - new bonds might make the {silk / fibres / molecule / protein} stronger (1) - larger R groups will make the protein insoluble (1) OR - larger R groups will make the {silk / fibres / molecule / protein} weaker (1) - because {fibres / molecule / protein} not held so closely together (1)	ACCEPT tertiary structure IGNORE quaternary structure ACCEPT named example of bond changing	(4)